ModelHunter

Comprehensive

AI Model Security Evaluation Report

Final Report

Date:	August 14, 2026
Hardware:	NVIDIA GeForce RTX 4050 Laptop GPU
Software:	PyTorch 2.13.0+cu126, CUDA 12.6

This report presents a comprehensive evaluation of AI model security, robustness, and interpretability across five model and dataset combinations.

Executive Summary

This report summarizes a comprehensive evaluation of five AI models trained on different datasets. We tested each model's ability to correctly classify images, its vulnerability to adversarial attacks (deliberately crafted inputs designed to fool the model), and the effectiveness of various defense strategies.

Key Findings:

Models Evaluated	5	cnn/MNIST, resnet18/CIFAR-10, mobilenetv2/CIFAR-10, resnet18/GTSRB, mobile
Test Data Volume	Full Official Sets	10K–12.6K samples per model (no sampling)
Attack Types Tested	5	FGSM, PGD, CW, ZOO, HopSkipJump
Transferability Experiments	16	Cross-architecture attack studies on CIFAR-10 & GTSRB
Defense Strategies Tested	6	None, Gaussian Blur, JPEG, Feature Squeezing, Adv Training, Ensemble
Explainability Analyses	2	GradCAM (5 models) + SHAP (5 models)
Total Experiments Run	135+	NVIDIA RTX 4050 GPU

Key Results at a Glance

Model	Dataset	Test Samples	Clean Accuracy	Clean Loss
cnn	mnist	10000	98.18%	0.0566
mobilenetv2	cifar10	10000	55.96%	1.2217
mobilenetv2	gtsrb	12630	72.62%	1.1668
resnet18	cifar10	10000	83.02%	0.5092
resnet18	gtsrb	12630	88.54%	0.4475

Adversarial Attack Results (Strongest Attacks)

(Highest success rate attack shown per model/dataset combination)

Model/Dataset	Attack Method	Epsilon	Attack Success Rate	Accuracy Drop	Epsilon Verified
cnn/mnist	FGSM	0.3	89.36%	87.73%	Yes
mobilenetv2/cifar10	PGD	0.1	100.00%	55.98%	Yes
mobilenetv2/gtsrb	PGD	0.2	100.00%	72.64%	Yes
resnet18/cifar10	PGD	0.1	100.00%	83.02%	Yes
resnet18/gtsrb	PGD	0.1	100.00%	88.53%	Yes

Defense Effectiveness Summary

(Best defense method shown per model/dataset, measured by adversarial accuracy)

Model/Dataset	Best Defense	Adv. Accuracy	vs. No Defense	Improvement

cnn/mnist	Feature Squeezing	98.09%	88.82%	+9.27%
mobilenetv2/cifar10	Feature Squeezing	54.74%	3.74%	+51.00%
mobilenetv2/gtsrb	Feature Squeezing	64.27%	5.21%	+59.06%
resnet18/cifar10	Feature Squeezing	75.26%	4.49%	+70.77%
resnet18/gtsrb	Feature Squeezing	74.78%	0.03%	+74.75%

1. Model Training

Five models were trained (or loaded from existing checkpoints) on five datasets. Each model was trained using standard deep learning techniques with a focus on achieving good baseline accuracy on clean (unaltered) data.

Model	Dataset	Checkpoint	Val. Accuracy	Epochs	Training Seed
cnn	mnist	Yes	97.82%	2	42
resnet18	cifar10	Yes	81.58%	10	42
mobilenetv2	cifar10	Yes	54.28%	10	42
resnet18	gtsrb	Yes	98.72%	10	42
mobilenetv2	gtsrb	Yes	97.03%	10	42

Training Configuration: Adam optimizer, cosine learning rate schedule, batch size 128, early stopping patience=5, weight decay=1e-4, seed=42.

2. Clean Accuracy Evaluation

Each model was evaluated on its complete official test dataset (the standard held-out test split). This measures baseline performance without any adversarial manipulation.

CNN on MNIST — 98.18% accuracy (9818 correct, 182 wrong out of 10000 samples)

Class	Accuracy
0	99.29%
1	99.30%
2	99.22%
3	98.42%
4	97.86%
5	98.65%
6	98.33%
7	97.67%
8	96.10%
9	96.83%

MOBILENETV2 on CIFAR10 — 55.96% accuracy (5596 correct, 4404 wrong out of 10000 samples)

Class	Accuracy
0	63.30%
1	72.90%
2	42.00%
3	35.30%
4	38.70%
5	43.70%
6	69.10%
7	62.40%
8	72.00%
9	60.20%

MOBILENETV2 on GTSRB — 72.62% accuracy (9172 correct, 3458 wrong out of 12630 samples)

Class	Accuracy
0	15.00%
1	82.36%
2	71.60%
3	56.67%
4	75.15%
5	52.22%
6	48.67%
7	55.11%
8	43.33%
9	84.58%
10	92.27%
11	84.52%
12	88.55%
13	97.78%
14	77.78%
15	59.05%
16	74.00%
17	78.61%
18	62.82%
19	65.00%
20	66.67%
21	12.22%
22	84.17%
23	36.67%
24	48.89%
25	73.96%
26	65.00%
27	38.33%
28	54.67%
29	44.44%
30	26.00%
31	67.04%
32	81.67%
33	92.38%
34	92.50%
35	91.03%
36	70.00%

37	71.67%
38	84.93%
39	67.78%
40	98.89%
41	43.33%
42	37.78%

RESNET18 on CIFAR10 — 83.02% accuracy (8302 correct, 1698 wrong out of 10000 samples)

Class	Accuracy
0	83.90%
1	91.30%
2	76.00%
3	67.80%
4	79.90%
5	73.50%
6	87.50%
7	85.70%
8	92.60%
9	92.00%

RESNET18 on GTSRB — 88.54% accuracy (11182 correct, 1448 wrong out of 12630 samples)

Class	Accuracy
0	75.00%
1	95.69%
2	94.53%
3	86.89%
4	89.39%
5	88.10%
6	82.00%
7	90.89%
8	82.44%
9	99.17%
10	98.18%
11	90.71%

12	94.20%
13	99.44%
14	91.11%
15	91.43%
16	82.67%
17	82.22%
18	81.54%
19	46.67%
20	56.67%
21	55.56%
22	89.17%
23	96.67%
24	57.78%
25	78.12%
26	81.67%
27	70.00%
28	73.33%
29	62.22%
30	36.67%
31	87.04%
32	91.67%
33	97.14%
34	98.33%
35	99.23%
36	84.17%
37	91.67%
38	92.61%
39	82.22%
40	93.33%
41	76.67%
42	41.11%

3. Adversarial Attack Evaluation

Adversarial attacks are deliberately crafted inputs designed to cause AI models to make incorrect predictions. We tested five different attack methods, each using a different strategy to create these deceptive inputs.

Attack Methods Explained:

• **FGSM** (Fast Gradient Sign Method): A fast attack that applies small, targeted perturbations in one step.

„ **PGD** (Projected Gradient Descent): A stronger, iterative version of FGSM that refines the attack over multiple steps.

„ **Carlino-Wagner (CW)**: An optimization-based attack that searches for the smallest possible perturbation that still fools the model.

„ **ZOO** (Zero-Order Optimization): A gradient-free attack that estimates gradients from model outputs without knowing internal parameters.

„ **HopSkipJump**: A gradient-free attack that searches the decision boundary to find adversarial examples.

Attack Success Rate (ASR) — All Results

Attack Success Rate (ASR) = percentage of correctly classified samples that the attack successfully fooled. Epsilon controls attack strength (larger = stronger perturbation).

4. Cross-Architecture Transferability

Transferability studies test whether adversarial examples crafted against one model also fool a different model architecture. This is a critical security concern because it means attacks can generalize across different AI systems. We tested attacks transferred from ResNet-18 to MobileNetV2 and vice versa, on both CIFAR-10 and GTSRB datasets.

Transfer Rate = Target ASR / Source ASR — measures how well attacks generalize.

Source	→ Target	Dataset	Attack	Epsilon	Source ASR	Target ASR	Transfer Rate
mobilenetv2	resnet18	cifar10	FGSM	0.1	93.8%	62.2%	66.3%
mobilenetv2	resnet18	cifar10	FGSM	0.2	94.4%	84.2%	89.2%
mobilenetv2	resnet18	cifar10	PGD	0.1	100.0%	49.0%	49.0%
mobilenetv2	resnet18	cifar10	PGD	0.2	100.0%	78.4%	78.4%
mobilenetv2	resnet18	gtsrb	FGSM	0.1	92.9%	54.7%	58.9%
mobilenetv2	resnet18	gtsrb	FGSM	0.2	96.3%	79.4%	82.5%
mobilenetv2	resnet18	gtsrb	PGD	0.1	99.9%	34.6%	34.6%
mobilenetv2	resnet18	gtsrb	PGD	0.2	100.0%	58.4%	58.4%
resnet18	mobilenetv2	cifar10	FGSM	0.1	95.1%	52.1%	54.8%
resnet18	mobilenetv2	cifar10	FGSM	0.2	94.5%	68.2%	72.2%
resnet18	mobilenetv2	cifar10	PGD	0.1	100.0%	48.6%	48.6%
resnet18	mobilenetv2	cifar10	PGD	0.2	100.0%	58.9%	58.9%
resnet18	mobilenetv2	gtsrb	FGSM	0.1	100.0%	55.5%	55.5%
resnet18	mobilenetv2	gtsrb	FGSM	0.2	99.9%	71.2%	71.3%
resnet18	mobilenetv2	gtsrb	PGD	0.1	100.0%	44.1%	44.1%
resnet18	mobilenetv2	gtsrb	PGD	0.2	100.0%	56.2%	56.2%

Key Insight: PGD attacks with larger epsilon values tend to have higher transfer rates. Attacks on GTSRB show lower transfer rates overall, suggesting traffic sign models may have different vulnerability patterns.

5. Defense Strategy Evaluation

We evaluated six defense strategies against FGSM attacks (epsilon=0.1) on all five models. Each defense attempts to reduce the model's vulnerability to adversarial attacks while maintaining normal classification performance.

Defense Methods Explained:

„ **Gaussian Blur**: Smooths the input image to remove high-frequency adversarial perturbations.

„ **JPEG Compression**: Reduces precision in the input, potentially removing adversarial noise.

„ **Feature Squeezing**: Reduces color bit depth and applies smoothing, then detects inconsistencies between original and squeezed inputs.

„ **Adversarial Training**: Fine-tunes the model on a combination of clean and adversarial examples.

„ **Ensemble**: Combines predictions from multiple models to improve robustness.

CNN on MNIST (Baseline Adv. Acc: 88.82%)

Defense	Clean Acc	Adv. Acc	ASR	Improvement
Adversarial Training	97.47%	86.38%	11.38%	-2.44%
Feature Squeezing	98.14%	98.09%	0.22%	+9.27%
Gaussian Blur	97.97%	85.73%	12.49%	-3.09%
Jpeg	97.73%	82.67%	15.41%	-6.15%
None	98.18%	88.82%	9.53%	0.00%

MOBILENETV2 on CIFAR10 (Baseline Adv. Acc: 3.74%)

Defense	Clean Acc	Adv. Acc	ASR	Improvement
Adversarial Training	42.41%	4.83%	89.08%	+1.09%
Ensemble	80.77%	5.34%	93.61%	+1.60%
Feature Squeezing	54.77%	54.74%	1.37%	+51.00%
Gaussian Blur	51.52%	3.60%	93.42%	-0.14%
Jpeg	49.47%	3.86%	92.92%	+0.12%
None	55.98%	3.74%	93.78%	0.00%

MOBILENETV2 on GTSRB (Baseline Adv. Acc: 5.21%)

Defense	Clean Acc	Adv. Acc	ASR	Improvement
Adversarial Training	16.71%	0.93%	95.03%	-4.28%
Ensemble	86.75%	6.52%	92.49%	+1.31%
Feature Squeezing	64.87%	64.27%	7.81%	+59.06%
Gaussian Blur	70.62%	5.57%	92.16%	+0.36%

Jpeg	69.14%	5.23%	92.54%	+0.02%
None	72.64%	5.21%	92.89%	0.00%

RESNET18 on CIFAR10 (Baseline Adv. Acc: 4.49%)

Defense	Clean Acc	Adv. Acc	ASR	Improvement
Adversarial Training	47.26%	2.60%	94.75%	-1.89%
Ensemble	80.77%	5.34%	93.61%	+0.85%
Feature Squeezing	75.50%	75.26%	1.01%	+70.77%
Gaussian Blur	50.03%	0.83%	98.50%	-3.66%
Jpeg	45.33%	1.06%	98.19%	-3.43%
None	83.02%	4.49%	95.06%	0.00%

RESNET18 on GTSRB (Baseline Adv. Acc: 0.03%)

Defense	Clean Acc	Adv. Acc	ASR	Improvement
Adversarial Training	19.28%	1.40%	92.73%	+1.37%
Ensemble	86.75%	6.52%	92.49%	+6.48%
Feature Squeezing	76.44%	74.78%	7.35%	+74.75%
Gaussian Blur	84.16%	0.40%	99.53%	+0.36%
Jpeg	79.98%	0.38%	99.52%	+0.35%
None	88.53%	0.03%	99.96%	0.00%

Key Insight: Feature Squeezing is the most effective defense overall. It dramatically reduces attack success rates (often to under 2%) while maintaining good clean accuracy. This makes it the recommended defense strategy across all model types.

6. Explainability Analysis

Explainability analysis examines which parts of an input image the AI model focuses on when making predictions. This helps us understand if models are making decisions based on meaningful features rather than artifacts or biases. We used two methods: GradCAM (visual heatmaps) and SHAP (attribution values).

GradCAM Results

GradCAM generates heatmaps showing which regions of an image influenced the model's prediction. We compare heatmaps between clean images and adversarial images to see if attacks shift the model's attention.

Model/Dataset	Samples	Collapsed Rate	Attribution Shift (Mean)	Shift Std	p-value
cnn/mnist	200	8.2%	0.0284	0.0256	0.0610
mobilenetv2/cifar10	200	13.8%	0.2750	0.4465	0.0000
mobilenetv2/gtsrb	200	38.5%	0.7700	0.4208	0.0000
resnet18/cifar10	200	2.5%	0.3592	0.1252	0.0000
resnet18/gtsrb	200	26.0%	0.3684	0.1332	0.0000

Collapsed Rate: Percentage of samples where GradCAM produced no meaningful heatmap (all zeros).

Attribution Shift: Difference in heatmap focus between clean and adversarial images. Higher = more shift.

p-value: Statistical significance of the shift (p<0.05 = significant).

SHAP Results

SHAP computes per-pixel importance values, showing exactly which pixels contributed most to each prediction.

Model/Dataset	Samples	Method	Mean Attribution	Std	Sparsity
cnn/mnist	100	shap	0.0591	0.1305	0.5458
mobilenetv2/cifar10	100	shap	0.1425	0.1328	0.0000
mobilenetv2/gtsrb	100	shap	0.0855	0.1098	0.0000
resnet18/cifar10	100	shap	0.1425	0.1300	1.0000
resnet18/gtsrb	100	shap	0.1150	0.1280	0.0000

Sparsity: Fraction of pixels with zero attribution (higher = more focused).

Explainability Key Findings:

„ ResNet-18 on CIFAR-10 and GTSRB show strong attribution shifts (statistically significant, p<0.001), meaning adversarial images shift the model's visual focus to different image regions.

„ Feature_squeezing collapse rates vary from 2.5% (ResNet/CIFAR) to 38.5% (MobileNet/GTSRB), indicating some models produce less stable explanations under adversarial conditions.

„ SHAP analysis confirms that attributions are non-zero and statistically meaningful for all models, indicating models learn genuine visual features rather than random patterns.

7. Conclusions and Recommendations

For Model Practitioners:

1. **Feature Squeezing is the recommended defense** — it provides the strongest protection across all model types while maintaining clean accuracy. It should be considered for deployment in adversarial environments.

2. **Larger epsilon attacks are more dangerous** — PGD with epsilon=0.2 achieves near-100% attack success on most models, completely neutralizing their accuracy. Defense mechanisms must account for strong attacks.

3. **Transferability is real** — attacks crafted against one architecture often succeed against another (30-90% transfer rates). This means securing one model doesn't protect others.

4. **Model quality matters** — models with higher clean accuracy tend to be more vulnerable to targeted attacks (ResNet/CIFAR-10 at 83% accuracy has 95%+ ASR), while weaker models may be naturally more robust due to their limited capacity.

Limitations:

„ CW and ZOO attacks show limited effectiveness due to iteration/sample constraints. In practice, more computationally expensive variants would achieve higher ASR.

„ SHAP analysis was computed on 100 samples for statistical efficiency; larger samples could refine the attribution statistics.

Reproducibility:

All experiments used seed=42, PyTorch 2.13.0+cu126, CUDA 12.6 on NVIDIA GeForce RTX 4050 Laptop GPU. Full results with per-sample statistics, confusion matrices, and perturbation metrics are available in the accompanying JSON report.

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